Force and Balance

PSIC's Transformative Concepts
Jan 2025 v1

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Introduction to Forces and Balance in Skiing

At PSIC, we embrace a first principles approach as the foundation of our educational philosophy. This approach starts with the science—objective, fundamental truths—and translates those principles into practical, relevant programming for ski professionals. A prime example of this philosophy in action is the concept of Force and Balance, universal elements that define how skiing works. These principles are integral to the PSIC's Technical Blueprint, a framework designed to observe, assess, and develop skiing effectively.

Force and Balance govern our interaction with the world on skis. The gravitational pull of the earth, the push of the snow, and the act of balancing with these changing forces make skiing possible. Rooted in Newtonian physics and biomechanics, these principles provide the scientific foundation for understanding the sport, offering a lens to explain and simplify its complexities.

These principles are why we move the way we do on skis and why skis are designed as they are—applicable across countries, organizations, disciplines, and physical abilities. With Force and Balance, skiing's complexities become simpler. They form the foundation of skiing mechanics, connecting every technical aspect of the sport back to these fundamentals. A solid understanding of Force and Balance makes all aspects of skiing make more sense.

Skiing is all about managing external forces to control our motion. Gravity pulls us down the slope, while we use our skis to engage with the snow in a specific way to generate forces that dictate speed and direction. Staying upright involves aligning our bodies with these external forces—balancing. *Force and Balance* are the two essential building blocks of skiing and represent PSIC's transformative concepts. Ultimately, any skier's action can be distilled down to its effect on these foundational elements. A solid grasp of these concepts will make you a better skier, teacher, and trainer.

Practical Applications:

Practical Application sections have been included. These provide opportunities to place the knowledge presented into a real-world, practical context. These additions challenge you to engage deeply with the material, interrogating the concepts and examining how they apply to your work as a ski professional. This process helps transform knowledge into *know-how*—the ability to apply information effectively in context-specific scenarios. By translating knowledge into actionable skills, these sections aim to build confidence in applying or relating the material to real-world teaching, making your instruction more impactful and relevant.

At the reader's discretion, it may be helpful to first read through the paper without engaging with the practical application sections. This approach can provide a clearer understanding of the subject matter before incorporating the detailed practical application components.

Understanding Force in Skiing

In skiing, force is simply a push or pull acting on the skier. We have two categories of Forces in skiing, internal and external forces. A fundamental concept in skiing is the interaction between the skier and the snow surface. Between the snow and the bases of our skis is where we feel the interplay and effect of these 2 forces. When we apply force to the snow through our skis, it pushes back—this pushback is the essence of skiing mechanics. This relationship aligns with Newton's Third Law of Motion: for every action, there is an equal and opposite reaction. This reciprocal force allows us to turn, stop, and control our movement on the mountain.

Gravity: The Driving Force

Gravity is a primary force in skiing. It holds us to the mountain and continuously pulls us down the slope, acting as the engine that drives our motion. As we descend, gravity imparts momentum to our motion. Changing speed and direction means changing momentum. Skiers manage momentum by manipulating how their skis interact with the snow.

Whether accelerating, slowing down, or at maintaining a constant speed, gravity is always at work on the skier, creating both the need and the opportunity to control speed and direction.

Managing External Forces for Speed and Direction

To control speed and direction, a skier manipulates how their skis interact with the snow so that the snow applies force at a specific angle and magnitude to achieve the desired effect. It's this force that guides the skier through a turn, controls their speed, or brings them to a stop, depending on the angle and direction of the push from the snow.

Skiers manage this interaction by tipping, twisting, and adjusting pressure on the skis, allowing the skis to exert force on the skier that affects their path down the hill – their motion. We refer to this motion as the skier's direction of travel, or *line of momentum*. For effective turning or slowing down, this push from the snow must be angled relative to the line of momentum.

Practical Applications:

There are countless examples of how managing external forces plays out in skiing. Imagine the experience of a new skier achieving their first successful speed control by positioning their skis in a wedge, creating resistance against forward motion. Or consider the excitement of their first intentional acceleration as they decrease that same wedge, reducing the slowing resistance—a thrilling moment.

Direction change, too, is a dynamic application of managing forces. For the novice skier, it might start by emphasizing one ski more than the other within the wedge, creating enough of a force that pushes them in a new direction. Meanwhile, the accomplished skier may use more athletic and refined movements, such as shifting the skis midair to position them for a powerful new turn

entry. Or they might use the terrain itself, positioning their skis to “bounce” off features, like the ball in a pinball machine hitting a bumper, to alter their trajectory quickly.

As skill progresses, so does the refinement in creating these forces. Consider the precision required to carve a turn—a skier adjusting their ski position and applying force just so, allowing the skis to arc without slowing. And on extreme terrain, like a steep couloir, expert skiers apply finely tuned timing and pressure directly opposing their line of direction to manage their momentum, riding the line between acceleration and deceleration with precision.

The mountain environment offers endless scenarios of managing external forces. Each scenario requires its own purpose to achieve the desired outcome. How many other examples can you identify where managing external forces influences skiing outcomes?

Internal Forces and Movements: The source of Control

External forces dictate the outcome of *what* happens with our motion, but it’s our internal forces— movements generated by our muscles—that allow us to manage and shape *how* this motion unfolds.

By continually adjusting our skis and posture through movement, we generate internal forces that create the necessary control and stiffness. Through these two strategies, we manage internal forces to meet the demands of external forces.

Practical Applications:

Internal forces enable skiers to position their skis and adapt their posture through movement, allowing them to create and manage the forces described in all of the examples above.

For a beginner, actions like sidestepping uphill, positioning the skis into a snowplow, or flexing the ankles to press forward in the boots—all rely on internal forces. Imagine a skier navigating variable terrain; they continually adjust their posture to absorb forces exerted by uneven snow. Rotating the legs to create a steering angle, aligning body segments to achieve angulation, or rolling the knees inward for greater ski grip—each of these movements is driven by internal force and stiffness through the linkage. Likewise, stiffening muscles to absorb the impact of a cliff jump or performing complex movement patterns, such as allowing the upper body to move into the new turn while the skis and legs continue along the current turn for a split second longer, demonstrate how internal forces are constantly engaged.

In each scenario, the skier relies on internal force and functional stiffness to position and adapt their body, using their movements to set up the prerequisite actions needed to extract external forces. Consider the range of internal adjustments required as terrain, speed, and intent change—each demanding a precise combination of movements, alignment, and timing. What other scenarios can you identify where internal forces play a critical role in skiing?

Balance

Let's now introduce *Balance*—the other half of the PSIC's core concepts and the essential partner to *Force* in skiing. Balance is essential in skiing, as it helps us manage external forces acting on us and enables control and stability during movement.

Simply put, balance in skiing is the result of a favorable relationship between the center of mass (COM) and the base of support (BOS). When the forces acting on our center of mass align with our base of support (BOS), we feel stable; we feel in balance. With the introduction of motion, so comes the requirement to manage momentum and turning forces. Achieving this balance involves continuously positioning the body to manage the forces acting on it – Dynamic balance. Dynamic balance provides skiers with the stability needed to execute precise mechanics and achieve desired outcomes. It's a dynamic process, almost like a dance with the mountain—an ongoing adjustment of movement and posture while in motion.

Practical Applications:

Beyond the basic understanding that it's hard to accomplish anything while struggling to stay upright and skiers tire quickly if they're constantly battling for balance..

Most ski pros experience a breakthrough when they realize that many balance issues aren't about basic positioning or stance; instead, they stem from students not finding a stable connection to balance against (grip). For a beginner, that very first straight run often ends with them leaning too far back or even falling backward, as they haven't yet sensed the force they need to balance against. The moment they engage a wedge, aligning themselves perpendicular to the slope, they create a resisting force they can feel pushing back at them—suddenly, the brain recognizes something solid to balance on, or against, providing stability.

Consider a new parallel skier who has that precarious moment as they start every new turn until they feel the skis push back. That pressure providing the safety their brain was seeking. Or think about an advanced skier setting up for a carve on a fresh groomer, only to find the outside ski doesn't push back because the snow hasn't had time to set after grooming, crumbling underfoot instead. Similarly, on icy terrain without perfectly sharp edges, skiers often lose confidence due to the lack of grip early in the turn. Without that resistance, it becomes difficult for the brain to commit to moving the COM further inside the turn. These balance issues all stem from the skier's inability to connect their balance system to something dependable.

Understanding when to sacrifice stability for mobility is another key insight. For example, we teach beginners to start in a snowplow not only because it simplifies steering but also because it provides a broad, stable base of support. As skiers progress and require greater mobility, we reduce the base of support, trading some stability for increased maneuverability. In powder, we narrow the stance further to create an enhanced unified platform.

Other balance-enhancing tactics include the stability and support of a well-timed pole plant—essential for skiing bumps or steeps. Dragging the inside pole can increase proprioception and improve upper body stability, enabling stronger leg turning and edging, while pulling the inside ski back helps engage and stabilize the hips.

The Interplay of Force and Balance

As skiers, Teachers and Trainers, the relationship between force and balance offers us more insight than the individual components alone. Understanding each component individually is essential, yet the true value lies in grasping their dynamic interaction. When we ski, we balance with the forces at play—hence the concept of Dynamic Balance.

The conventional notion of balance as simply an up-and-down alignment with gravity no longer applies when more substantial turning, slowing and terrain induced forces are involved. Without accounting for the external forces that skiing introduces—and adjusting movement to work with them—a skier would lack the stability needed to execute proper mechanics. Simply put, they wouldn't stay upright.

Recognizing and generating the forces needed from the skis to achieve a specific purpose, and understanding how these forces act on the skier, and the skier's ability to align with them, captures the essence of interplay between Force and Balance. This insight not only enhances personal skiing mechanics but also strengthens a professional's ability to recognize and teach mechanics of skiing to others effectively.

Novice skiers experience minimal forces, but as speed and slope increase, these forces grow significantly. Greater turning forces require a larger lateral range of movement, with increasing inclination angles needed to manage the heightened forces. This progression demands a more advanced level of dynamic balance and body performance. As skiers turn, their bodies naturally shift inside the arc to maintain balance—the greater the force, the more pronounced this movement becomes

A skier's stability—or intentional disruption of it, such as moving further inside the turn or influencing momentum into a new turn—is anchored off an axis that aligns not simply with gravity but also accounts for turning forces (centripetal force). This is felt as centrifugal force—the sensation of being pulled to the outside of the turn. To maintain balance, skiers must lean into the turn, aligning the forces acting on their body so they pass through or near their BOS. This controlled instability allows for smooth transitions into and out of turns.

Advanced skiers continuously predict, anticipate, and prepare for large external forces by making preparatory movements or adjustments, positioning themselves to manage these forces effectively. You might say they "give to get."

Balanced skiers flow, moving smoothly with the forces acting on them and transitioning intentionally from turn to turn. Skiing is a game of creating and managing forces to push us around while maintaining balance with them.

Practical Applications:

An aggressive hockey stop is a prime example of the force and balance interplay. Visualize the skier in this maneuver: the rapid deceleration requires a distinct change in the relationship between the feet and the hips, first shifting inside and down and then back up. At peak deceleration, the hips align significantly inside the skis to counter the stopping force and then immediately return above the skis as that force diminishes.

Think of the kid ripping into a hockey stop, slamming the skis sideways with a massive spray of snow. Their hip is nearly on the snow, aligning with the forces at play—but if their timing isn't precise, the moment the stopping force stops, they end up sitting down, often laughing at how cool the move just felt.

Not every instance of force and balance interaction is as striking as a hockey stop; often, this relationship is more subtle. Take, for example, the very first snowplow. As a beginner creates a braking force with the skis, they instinctively shift their hips slightly back to maintain balance, preventing a forward pitch. Or consider the subtle lean an intermediate skier adopts into a turn as lateral forces push from the outside—small adjustments like these reveal the continuous dance of balance, alignment, and force in skiing.

Versions of force alignment mechanics appear everywhere in skiing once you know what to look for. Consider how the contact force—where the length of the ski base meets the snow—always acts perpendicular to the slope. Aligning with this force is crucial as a skier initiates a turn, adapting to the hill's increasing steepness. Just as one naturally projects forward onto a moving surface, like an airport walkway or escalator, to match its speed, a skier must adjust their alignment to stay in sync with the changing slope and acceleration as the skis are directed downhill. This can be achieved through extending the knees or pulling the feet back to maintain an optimal force-balance relationship. While the choice of tactic may vary by situation, the underlying goal remains the same: to optimize the force balance relationship for the purpose, in this case keeping the skier positioned perpendicular to the slope.

How about, shifting the feet forward before hitting a bump, transitioning from groomed terrain into soft snow, or absorbing the impact from a cornice drop. All these examples illustrate how balance adjustments, whether anticipatory or reactive, set up the essential force-balance relationship.

Every turn contains elements of this force-balance dynamic, with faster speeds and tighter radius demanding the hips to be positioned increasingly inside the turn. Think on how expert skiers are adept at allowing their COM to fall just slightly ahead of the ski path that continuously catches them. Consider the athletic mobility and force balance interaction observed in a World Cup GS Stivot turn where the racer rapidly falls inside, often hip to snow and arrests the fall in a one-shot edging action (drift and set). Very much a curvilinear (opposed to linear) version of the hockey stop, with as little skid and as much carve as possible as the skis bite.

The interplay between external and internal forces offers extensive options for initiating a new turn. Directing our mass into the new direction can be accomplished by mechanics such as softening the outside leg, extension of the uphill leg, or combinations of independent leg actions. Techniques like intentionally slowing the skis and feet via a check turn in the steeps, tightening the radius, allowing the momentum of the upper body to yield momentarily to the forces pulling us outward while the legs and skis continue to complete the turn for a moment longer—all exemplify this dynamic. Combining any of these actions, along with timing, showcases the incredible versatility and control within the Force-Balance interplay.

The range of options and movements available to an expert skier is astounding. By understanding the dynamic relationship between Force and Balance, it becomes easier to

reason through what needs to happen, identify options to achieve it, and teach those skills effectively—bringing clarity to what might otherwise seem complex.

Pressure Control: Managing force for Optimal Performance

As skiers, our frame of reference is internal. We are sensory creatures, equipped with an innate ability to feel and perceive our body's posture, position, and movements relative to the forces acting on us - proprioception. This internal compass—or gyroscope—provides us with a sense of where our bodies are in space and how external forces act upon them. This awareness enables us to dynamically respond to these forces, allowing us to manage them in motion and optimize performance—our dance with the mountain. In skiing, managing these external forces is often referred to as pressure control.

The force pushing back from the snow creates a sensation of weight, or pressure (*pressure = force over a given area, in this case our skis*), between the skis and the snow. Within our base of support, there is a theoretical point where all forces from the snow converge and act on the skier—this is known as the *Center of Pressure* (COP). It serves as the focal point, a point of application of force exerted by the snow. Adjusting the COP changes how the snow pushes back, allowing greater control over the skis' response and performance. For example, shifting pressure toward the front, the tail, or from one ski to the other directly affects how the skis behave on the snow, and as discussed above, how the snow pushes on the skier.

As we ski, we are continually adjusting the interaction between our skis and the snow to create and manage force. We interpret and manage these forces based on how we feel them in real-time. Balancing against these forces involves aligning our COM and BOS to achieve the desired effect for any given purpose. This feeling, this perception of forces acting on us, creates opportunities for adjustment—allowing us to refine our control, direct the effects, and influence our motion with precision. Make no mistake, expert pressure control is the mark of an expert skier.

Maintaining a steady, balanced mass while allowing the legs to actively respond to the continuous variations in the snow is possible only because skiers can feel these forces in the first place. Anticipatory movements stem from our interpretation of upcoming changes—our internal sense of 'how this is going to feel' and how to prepare for it. These reactive and anticipatory adjustments are then immediately confirmed by how they feel in action.

Creating focused experiences through varied terrain and surface changes—terrain play—helps skiers improve their ability to interpret and execute instinctual actions in different situations (*Instinctual movements are learned anticipatory actions stored in the brain as muscle memory - engrams)*. These exercises develop skiers "blueprints" for physical actions, formed through repetition and practice. When a skier repeatedly practices a specific movement, such as shifting weight into a turn or preparing for external forces, the brain encodes these actions. Over time, these stored patterns allow the skier to perform movements automatically, without conscious thought, enabling quick and precise responses to changing conditions on the slopes.

Connecting outcomes to sensations, recognizing when they'll occur, and reflecting on them to reinforce these sensations is a powerful teaching technique that enhances students' learning,

safety, and enjoyment. This practice develops the neural pathways needed to refine their *base representations*—the mental models that underpin decision-making and motor control.

Practical applications:

Let's unpack a few examples to illustrate how skiers execute maneuvers by creating, sensing, and reacting to force—this is pressure control in action. Developing, connecting, and honing pressure control is arguably the most functional skill we can impart to our students.

Consider momentum: a skier has an internal awareness of their motion, a sense of their speed, and an understanding of the force required to stop or change its direction. In the case of a hockey stop, they need an exceptionally strong force applied directly against their line of momentum. They also have an internal sense of where to position and align themselves to manage this force. As they feel the impact and its immediate effect on their momentum, they actively adjust their posture in response, recognizing in real time how their momentum is diminishing. Creating the appropriate force to arrest their momentum, while dynamically adjusting their posture to balance against this force, ultimately determines a successful stop—standing upright rather than ending up on the snow.

As an instructor, you already know what the skis must do to achieve this maneuver, the most effective movements for the purpose, and what the skier will feel in the process. Guiding students to connect with that feeling is essential for fostering improvement and building autonomy. Intentionally working in varied situations with focus targeted to helping them notice the sensations, both when things go well and when they don't.

Examples of how to develop pressure control are boundless. Let's look at edging—how much the skis move sideways as they move forward, or the amount of drift. For ski professionals, working on a skier's ability to edge and de-edge is a daily practice. Hopefully, you're including de-edging in that work, too. From the learner's perspective, how do they know they're improving? They feel it: they experience more grip, more bite, and a stronger pressure from the skis. A skier's ability to control the level of grip—adding or reducing it—comes down to their sensation of that grip.

Think about the skill it takes to hold a ski on the edge so that it's almost carving, just barely drifting, or to adjust from a drift to a pure carve. How do you know in the moment that you're achieving this? It's through the sensation—the feel of it. Sure, you can check your tracks at the bottom to confirm, but in the moment, it's your feel that enables you to optimize ski performance.

Prime students with specific focus on the sensations of their actions and the resulting outcomes. Set up situations that encourage them to reflect and build a connection to how things that feel: 'When you did that, what happened? What did you feel?" Help them connect feelings to results.

Here are some examples to get you started: using inside leg flexion to control inclination, adjusting edge angle, various movements or combinations for initiating a new turn, maintaining contact with the snow in expert short turns, or employing angulation to increase edge engagement without increasing lean. How many other examples can you bring to life in your teaching?

Effective Skiing: A First Principles Approach

Approaching skiing from a first principles perspective means understanding the interplay of Force and Balance. Gravity propels us downhill, and we manipulate our skis to manage forces that control speed and direction, constantly balancing against these shifting forces. This foundational knowledge enhances both skiing and teaching by providing a clear, science-based view of skiing mechanics.

Everything we present here stems from scientific fundamentals, which then translate into real-world applications of our craft. By appreciating what Force and Balance reveal, instructors gain a simplified, insightful lens into the mechanics of skiing. These principles encourage practitioners to reflect, reason, and plan with clarity. Force and Balance lie at the core of the PSIC's technical philosophy for a reason: they provide the essential intelligence needed to build expertise, adaptability, and consistency across all levels of skiing

Reference material: The PSIC's technical progression resource explores Force and Balance in relation to purpose-based outcomes in skiing. For further reading and deeper insights into this topic, please refer to these materials.